Fall 2023 CSE4070 Operating Systems

Midterm Exam - 2023.10.26 (Thursday)

1. True or False (30 points – 2 points each)

(1) In a uni-processor system, it's possible to have multiple processes in the running state.

(2) Response time refers to the duration between the submission of a request and the generation of the initial response.

(3) The convoy effect describes a situation in which both the CPU and I/O device resources are busy.

(4) In Named Pipe (FIFO), communication is uni-directional and has no parent-child relationship.

(5) System calls are implemented using an instruction which generates an interrupt.

(6) The multilevel feedback queue scheduling algorithm gives preference to processes with longer CPU bursts.

(7) A Round Robin (RR) scheduler produces results similar to a FIFO scheduler when the time quantum is shorter.

(8) A Round Robin (RR) scheduler with longer time quantum performs poorly on turnaround time than a RR scheduler with shorter time quantum.

(9) Processes in User mode can access all the resources on the machine.

(10) An OS that performs time-sharing can reduce the response time of each process by increasing the length of the time quantum.

(11) A trap instruction such as "**int n**" can raise the privilege level to kernel mode.

(12) One of the solutions to prevent another interrupt from occurring during interrupt or trap handling is to disable interrupts during interrupt processing.

(13) Cache coherence is a protocol designed to ensure the consistency of the contents of caches across multiple different CPUs.

(14) The entity performing bus snooping is the cache controller of each CPU.

(15) A multi-processor scheduler that considers cache affinity resolves the load balancing issue among CPUs.

2. Short Answer Questions (15 points – 3 points each)

(1) List any 5 of the attributes that an OS maintains to keep track of information about a process.

(2) List the reason why MLFQ scheduling algorithm favors an interactive job.

(3) List 2 reasons why the scheduling of processes and threads on a multi-processor system is more complicated than scheduling them on a uni-processor system.

(4) Typically, a process has three states (except "new/start" and "terminated/end". Draw a process state transition diagram, connect each state with edges, and provide detailed explanations for each node and edge.

(5) Imagine having **n** processes in the ready queue waiting to be scheduled on a single processor. How many distinct schedules are possible? Please provide a formula in terms of **n**.

3. Time Quantum in RR Scheduling (12 points – 6 points each)

(1) Consider **n** processes sharing the CPU in a round-robin (RR) fashion, making the following assumptions:

- Each context switch takes **s** milliseconds.
- Each process is guaranteed to have its turn on the CPU at least every **t** seconds.

Determine the largest quantum size, **q**, using a formula involving s, t, and n. Please provide a justification for your answer.

(2) Please fill in the blanks. Assume RR scheduling and justify your answer.

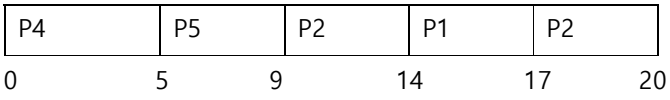
If there are **n** processes in the ready queue and the time quantum is **q**, then each process gets at least (①) CPU time in chunks of (②) time units. Each process can wait for a maximum of (③) time units until its next time quantum. Context switching overhead is ignored.

4. Process Scheduling (25 points) Suppose the following processes arrive for execution at the times indicated below. Each process will run for the listed amount of time (CPU burst is given in milliseconds) with the specified priority.

Process	Arrival Time	Burst Time	Priority
P1	6	6	2
P2	3	8	5
P3	1	10	2
P4	0	5	4
P5	5	6	1

1) Provide three Gantt charts illustrating the scheduled execution of these processes using the following algorithms:
   ① Preemptive Priority Scheduling (where a smaller priority number implies a higher priority).
   ② Non-preemptive Shortest Job First Scheduling.
   ③ Round Robin Scheduling (with a quantum of 6 ms).
   Assume FIFO scheduling for making decisions when two processes have the same priority."

(Example for Gantt chart)



2) What is the average turnaround time for each of the scheduling algorithms? Which of these scheduling algorithms results in the minimum average turnaround time?

5. MLFQ (18 points) This problem pertains to multi-level feedback queue (MLFQ) scheduling. However, this MLFQ differs slightly from what we learned in class. The following assumptions are made:

- There are two queues (Queue A, Queue B), and each queue operates with its own scheduling algorithm.
- Queue A employs a round-robin (RR) with a time quantum of 3, while Queue B operates with FCFS.
- The CPU dispatches processes from both queues using a weighted round-robin approach (2:1). In other words, the CPU dispatches two processes from Queue A and then one process from Queue B for execution, repeating this pattern continuously.

1) Four processes (P1, P2, P3, P4) have currently arrived in the ready queue in this order. P1 and P2 are in the Queue A, and P3 and P4 are in the Queue B. Processes do not move between Queues. Each process has a CPU burst time of 5 seconds. We assume that each process does not perform any I/O. Draw a Gantt chart.
2) What is the total wait time for each process in the ready queue.

-------- End of the Exam -------